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Ch.8 Plant Physiology

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

SECTION 1: MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. What is the evaporation of water from leaf surfaces, mainly through stomata?

Answer: (a) Transpiration — Transpiration is water loss as vapour, chiefly through stomata.

2. Through which structure does most transpiration occur?

Answer: (d) Stomata — Stomatal pores are the main route of water-vapour loss.

3. What does the TACT theory primarily explain?

Answer: (b) Transport of water in plants — Transpiration-pull/cohesion-tension theory explains upward water movement in xylem.

4. Which is not a function of xylem?

Answer: (c) Transport of food — Xylem transports water/minerals and gives support; food transport is phloem's role.

5. Which xylem element has perforated end walls?

Answer: (a) Vessel — Vessel elements have perforation plates, forming continuous tubes.

6. Low-temperature stimulation of flowering in biennials/perennials is called what?

Answer: (c) Vernalization — Vernalization is cold-induced flowering competence.

7. Plants adapted to dry conditions are called what?

Answer: (a) Xerophytes — Xerophytes have adaptations to reduce water loss and store water.

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8. When sugar content increases in a cell, what happens to water potential?

Answer: (b) Drops — More solute lowers (makes more negative) the water potential.

9. Food transport in higher plants occurs through what?

Answer: (b) Sieve tubes — Phloem sieve tubes transport organic solutes.

10. Which hormone inhibits stem and root growth?

Answer: (b) Ethylene — The textbook states ethylene inhibits stem and root growth; ABA also inhibits growth under stress but is not an option.

11. In floating hydrophyte leaves, stomata are mainly found where?

Answer: (b) Upper epidermis — The upper surface is exposed to air, so stomata occur there.

12. Mesophytes are adapted to which environment?

Answer: (a) Moderate environments — Mesophytes grow in conditions with moderate water availability.

SECTION 2: SHORT QUESTIONS

1. Differentiate between macronutrients and micronutrients.

Macronutrients	Micronutrients
Amount needed: Large amounts.	**Amount needed:** Trace amounts.
Examples: C, H, O, N, P, K, Ca, Mg, S.	**Examples:** Fe, Mn, Zn, Cu, B, Mo, Cl.
Role: Major structural and metabolic roles.	**Role:** Enzyme cofactors and regulatory roles.

2. What is water potential?

Water potential (Ψw) is the potential energy of water relative to pure water. It determines the direction of water movement: water moves from higher (less negative) to lower (more negative) water potential. In plant cells, Ψw = Ψs + Ψp (solute potential + pressure potential); pure water has Ψw = 0.

3. What are the three pathways for water movement between plant cells?

Apoplast pathway (cell walls and intercellular spaces), symplast pathway (cytoplasm of adjacent cells via plasmodesmata), and vacuolar pathway (crossing cell membranes, cytoplasm, tonoplasts and vacuoles).

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4. Differentiate between hypertonic and hypotonic solutions.

Hypertonic	Hypotonic
Solute concentration: Higher than cell sap.	**Solute concentration:** Lower than cell sap.
Water movement: Water leaves cell.	**Water movement:** Water enters cell.
Effect: Plasmolysis/shrinkage.	**Effect:** Turgidity/swelling.

5. What are halophytes?

Halophytes are plants adapted to salty environments. They may exclude salt, excrete salt through glands, store salt in vacuoles or shed salty leaves.

6. Differentiate between long-day and short-day plants.

Long-day plants	Short-day plants
Flowering: Flower when day length exceeds a critical value.	**Flowering:** Flower when day length is below a critical value.
Night requirement: Short night.	**Night requirement:** Long uninterrupted night.
Examples: Henbane, tomato, garden pea, maize, spring wheat/barley.	**Examples:** Cocklebur, chrysanthemum, soybean, tobacco, strawberry.

7. Write the phases of plant growth.

Formative phase (cell division at meristems), elongation phase (cell enlargement), and maturation/differentiation phase (cells become specialized).

8. Differentiate between vernalin and florigen.

Vernalin	Florigen
Stimulus: Cold treatment (vernalization, about 4°C).	**Stimulus:** Photoperiodic induction/biological clock.
Site: Shoot apex of mature stem or seed embryo.	**Site:** Leaves.
Role: Hormone that induces vernalization and flowering competence.	**Role:** Florigen hormone travels through phloem to floral buds to initiate flowering.

9. Differentiate between thigmotropism and geotropism.

Thigmotropism	Geotropism/gravitropism
Stimulus: Touch/contact.	**Stimulus:** Gravity.
Example: Tendrils coiling around support.	**Example:** Roots grow downward, shoots upward.

10. How is intercalary meristem different from apical meristem?

Intercalary meristem	Apical meristem
Location: Between mature tissues, e.g., bases of internodes/leaves in grasses.	**Location:** Tips of roots and shoots.
Duration: Temporary.	**Duration:** Perpetual growth zones.
Role: Regrowth/elongation of internodes, leaves and flowers.	**Role:** Primary growth in length.

SECTION 3: LONG QUESTIONS

1. Describe osmoregulation in hydrophytes and halophytes.

Hydrophytes

- **Freshwater hydrophytes:** live in hypotonic surroundings, so water continuously enters. They expel excess water through hydathodes or vacuoles, actively absorb ions such as K^+ and Ca^{2+}, and have thin/absent cuticles and reduced roots. Examples: water lily, lotus, tape grass.
- **Marine hydrophytes:** live in saline (hypertonic) water. They excrete excess salts through salt glands, synthesize organic solutes (proline, glycine betaine, sugars) to retain water, and may have thick cuticles and halophytic traits. Examples: seaweeds.

Halophytes

Halophytes grow in saline soils rich in NaCl, $MgCl_2$ or $MgSO_4$. They tolerate high salt by succulent leaves/stems that dilute salt, salt exclusion/excretion, and sometimes leaves modified into spines. Marshy seashore halophytes form mangrove/tidal woodland. Examples: sea lavender, mangroves.

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2. Describe physiological adaptations of plants to extreme conditions, including membrane/protein adjustments to temperature.

High temperature

- Plants use evaporative cooling (transpiration), but hot/dry weather forces stomata to close.
- Cells synthesize heat-shock proteins that surround and stabilize enzymes and other proteins, preventing denaturation.

Low temperature

- Cold makes membrane lipids crystallize and reduces fluidity; plants increase the proportion of unsaturated fatty acids to keep membranes fluid.
- Cold also affects membrane protein structure; adaptation requires time, so rapid chilling is more stressful than gradual cooling.
- Cold-region plants adjust cell solutes so the cytosol supercools without ice formation; ice may form in cell walls but not in the protoplasm, which would perforate membranes and kill cells.

Thus membrane lipid composition and protective proteins are key to surviving temperature extremes.

3. What is the role of meristems in plant growth?

Meristems are regions of actively dividing cells that produce new cells for growth.

Meristem	Location	Role
Apical	Tips of roots and shoots.	Primary growth: increase in length; perpetual growth zones.
Intercalary	Bases of internodes/leaves (e.g., grasses).	Temporary; regrowth and elongation of leaves/internodes.
Lateral (vascular cambium)	Between xylem and phloem.	Secondary growth: produces secondary xylem and phloem, increasing girth.
Lateral (cork cambium)	Outer layer of stems/roots.	Produces cork cells that replace epidermis and form protective bark.

Primary growth proceeds through zones of cell division, cell elongation and cell differentiation. Secondary growth increases thickness in woody plants.

4. Describe the mechanism of opening and closing of stomata.

Starch–sugar hypothesis

In light, guard cells photosynthesize and produce sugars, lowering water potential; water enters by osmosis, guard cells become turgid and the stoma opens. At night, sugars convert to starch, water leaves, guard cells become flaccid and the stoma closes.

K^+ influx hypothesis

> Light/blue light activates H^+ pumps → K^+ actively enters guard cells → Water follows by osmosis → Guard cells become turgid → Stoma opens

> Darkness or water stress (ABA) → K^+ leaves guard cells → Water follows out → Guard cells become flaccid → Stoma closes

Abscisic acid (ABA) released during wilting stops K^+ pumping, overriding light/CO_2 signals and closing stomata to conserve water.

5. Explain photoperiodism and its influence on flowering, including phytochrome.

Photoperiodism is the response of plants to the relative lengths of day and night. It is really the length of the dark period that is critical.

Short-day plants	Long-day plants	Day-neutral plants
Flower when nights are longer than a critical length; a light break in the night prevents flowering. Examples: cocklebur, chrysanthemum, soybean, tobacco.	Flower when nights are shorter than a critical length; a light break can promote flowering. Examples: henbane, tomato, garden pea, maize, spring wheat.	Flower regardless of photoperiod. Examples: cucumber, cotton, sunflower.

Phytochrome exists as P_{660} (Pr, absorbs red light) and P_{730} (Pfr, absorbs far-red light). Red light converts Pr → Pfr; far-red light/darkness converts Pfr → Pr. The rate of Pfr reversion in darkness acts as a biological clock measuring night length, ultimately triggering florigen production in leaves, which travels through phloem to floral buds.

The textbook's phytochrome paragraph contains an internal inconsistency about Pfr levels in short-day plants; the standard interpretation is that Pfr inhibits flowering in short-day plants and promotes it in long-day plants.

INQUISITIVE QUESTIONS

1. Explain the hypothesis for stomatal opening and closing.

Two hypotheses are given. (1) Starch–sugar hypothesis: in light, guard cells make sugars, water enters and they become turgid (open); at night sugars become starch, water leaves and they close. (2) K^+ influx/proton-pump hypothesis: H^+-ATPase pumps H^+ out, K^+ enters, water follows osmotically, turgor rises and the stoma opens; reverse ion movement causes closure. ABA promotes closure during water stress.

2. How do carnivorous plants supplement nutrients despite being autotrophs?

They photosynthesize for carbon but grow in nitrogen-poor soils. Traps capture insects; digestive enzymes absorb nitrogen and minerals, supplementing nutrient uptake.

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3. How do parenchyma and sclerenchyma provide support?

Parenchyma provides support through turgor pressure in soft tissues. Sclerenchyma has thick lignified walls and provides rigid mechanical support in mature regions.

4. How do annual rings depict climatic variability?

Each ring represents one year's secondary growth. Wide rings indicate favourable wet/warm seasons; narrow rings indicate drought/cold stress. Ring patterns therefore record past climate.

5. How does pressure-flow theory explain sugar movement?

Sucrose is actively loaded into sieve tubes at sources, lowering water potential; water enters from xylem, raising turgor/pressure potential. At sinks, sugar is unloaded, water leaves and pressure falls. The pressure gradient drives bulk flow of sap from source to sink.

The textbook (p191) says sucrose loading "increases water potential" at the source; in standard terms solute lowers water potential while pressure potential rises. The pressure gradient is what drives flow.

6. What strategies would you use to induce flowering in a plant?

Identify whether it is short-day, long-day or day-neutral; manipulate light/dark periods, provide vernalization if required, use appropriate temperature, and apply gibberellins where they can substitute for the required stimulus.

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